INCH-POUND

A-A-54625
15 April 1992

COMMERCIAL ITEM DESCRIPTION

CUTTER, ORTHOPEDIC CAST, Stryker, 110 Volt, AC-DC

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-C-36625 which is cancelled.

This Commercial Item Description covers an electric motor driven saw, with high speed oscillating drive, suitable for cutting surgical plaster casts.

Salient characteristics:

Style, design, and dimensions. Shall be an electric motor driven cast cutter, suitable for cutting cast on the human body without damage or injury to patient. Cast cutter shall be a non-rotating, high frequency, oscillating circular saw complete with plug adapter, wrench, one 2 inch and one 2-1/2 inch diameter circular blades. Style, design and dimensions shall be in accordance with the latest revision of DPSC Drawing 24160.

Motor and switch. Motor shall operate on either 110 volt direct current or 110 volt, single phase, 60 cycles per second, alternating current. Motor shall be fan cooled, permanently lubricated, universal type, rated at 0.1 horsepower at 10,000 revolutions per minute (RPM) continuous. Brush holders shall be protected by plastic plugs threaded into and flush with cast cutter housing. Shall have "off" and "on" switch mounted in housing. Motor and switch shall comply with the requirements of UL Standard 544 for continuous rated universal motors for electric tools.

AMSC N/A FSC 6515

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Power cable and plug. Cast cutter shall be furnished with a 3-wire, electrical power cord, 9 feet minimum length, and a 3-prong plug in accordance with ANSI Standard C72.11. The power cable shall be provided with either a spring type or an integrally molded component to assure strain relief of power cable at the housing end.

Blade arbor. The centerline of the blade arbor shall be offset 1 ± 1/8 inch from the centerline of the motor shaft, as shown on drawing, to provide blade in a safe and convenient position for cast cutter operation. The arbor shall be supported in bearing sleeves on each side of mechanism (open link) that transmits rotating motion into oscillatory motion. Blade arbor shall be designed to provide a proper and secure fit for the blades specified in this specification under "Blades".

Center of balance. With supply cable supported or detached, the center of balance of the cast cutter shall be 6-5/16 ± 1/2 inches from the front side of the mounted 2-1/2 inch diameter blade (measured parallel to the longitudinal axis of the housing).

Drive mechanism The rotary power of the motor shall be transmitted into an oscillatory action, at the blade end, by means of a cam and open link mechanism. The cam shall be equipped with a roller bearing. The open link mechanism shall be mechanically fastened to the offset blade arbor to suitably transmit the required oscillation rate. Gear transmission design shall not be acceptable.

Housing. The working mechanism shall be enclosed by a compact, rigid and thoroughly protected corrosion-resistant housing. Housing shall provide intake and outlet ports for oscillating mechanism. The housing, except for the ventilating openings, shall totally enclose all stationary and rotating electrical parts. All external fastening devices shall be recessed into the housing. Contours of mating housing parts shall match, with no sharp corners or edges. Housing components shall be accurately fitted with adequate oil seals and gaskets to prevent oil leakage.

Blades Each cast cutter shall be furnished with one 2 inch and one 2-1/2 inch diameter circular blade. (See Note 2).

Cast cutter weight. The cast cutter, including power cable, 3-prong plug and 2-1/2 inch diameter blade, shall not exceed 4-1/2 pounds (less carrying case, wrench and remaining blades).

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Fastening devices. All screws, pins, bolts and similar parts shall be installed with means of preventing loss of tightness.

Wrench. Each cast cutter shall be furnished with a wrench, as shown on the drawing, for attaching blades to arbor. Wrench shall be fabricated from corrosion-resistant metal.

Cast cutter performance. With input voltage set at 115 volts, the unit shall be capable of meeting the following performance characteristics:

Characteristic	Requirement
No load oscillation (strokes per minute (Free running)	17,000 minimum
Full load oscillations per minute under full load current of 1.2 amperes. . . .	13,000 minimum
No load current (Free running).	1.2 amperes maximum
Temperature rise of motor housing . . . . . during 5 minute operating period, under full load current of 1.2 amperes (when operated from a cold start)	40 Degrees Fahrenheit maximum over ambient temperature
Operation on duty cycle . . . . . breakdown or failure, under full load current of 1.2 amperes	100 hours minimum

NOTE: This test may be waived upon request by the contractor, when the contractor has previously furnished items in accordance with the requirements of this specification and performed this test, and there has been no major design change of motor and drive mechanism.

Cast cutter compatibility. Shall be compatible with vacuum cleaner specifically designed to collect and dispose of plaster dust during cutting or trimming orthopedic procedures by means of accepting the male connecting piece located on the distal end of the vacuum cleaner hose, as well as the electric power plug.

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Instructions for use. Each cast cutters shall have instructions for use and a list of replacement parts for the entire unit.

Identification marking and labeling. Cast cutter shall be legibly marked with the contractor's name or trademark. Foreign manufactured cast cutters shall bear the name of the country of origin.

Workmanship. The cast cutters shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One cutter consisting of one wrench, two blades, and carrying case, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as specified herein. Unit of product for examination purposes shall be one cast cutter. Sampling shall be Inspection level II with an acceptable quality level (AQL) of 1.0.

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Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one cast cutter. Sampling shall be Inspection level S-3 with an AQL of 0 for performance test.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more text characteristics.

Motor test. Each sample cast cutter motor shall be tested to determine compliance with UL Standard 544 requirements for continuous rated universal motors for electric tools:

Continuity of grounding connectors.

Dielectric strength.

Current input.

Current leakage.

Performance test. Sample cast cutters shall be tested for a least the following requirements:

Amperes at no load on 115 volts, 60 cycles, alternating current.

Spindle speed, oscillations per minute at no load.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

NOTE 1: The following National Stock Numbers are covered by this document:

NSN	ITEM IDENTIFICATION
6515-00-323-4510	CUTTER ORTHOPEDIC CAST, 110 VOLT, AC-DC.
6515-01-206-6028	CUTTER ORTHOPEDIC CAST, 110 VOLT, AC-DC.

NOTE 2: The blades shall be in accordance with applicable specifications. For 2 inch blade, NSN 6515-00-323-4515 and the 2-1/2 inch blade, NSN 6515-00-323-4520.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6515-4800

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54625	15 April 1992

3. DOCUMENT TITLE
CUTTER, ORTHOPEDIC CAST, Stryker, 110 Volt, AC-DC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: MSE	(1) Commercial (215)737-2870	(2) AUTOVON 444-2870
c. ADDRESS (Include Zip Code) 2800 S. 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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